Automate Banking Website

Using Agile Methodology

1. Abstract

An Automate Banking Website is a digital platform that allows customers to manage their bank accounts, perform transactions, and view financial records through the internet. With the rapid growth of digital finance, banks require secure, scalable, and user-friendly systems to meet customer demands. This project focuses on developing an Automate Banking Website using the Agile Software Development Methodology, which emphasizes iterative development, customer feedback, and continuous improvement.
Traditional banking software development models are often slow and difficult to modify once development begins. Agile methodology overcomes these limitations by dividing the project into small development cycles known as sprints. Each sprint delivers a functional module such as user authentication, fund transfer, balance inquiry, or transaction history. Feedback is collected after every sprint to improve the system.

The Banking website allows users to register, login securely, deposit money, withdraw funds, and transfer amounts to other accounts. Admin users can manage customer accounts, monitor transactions, and handle security alerts. Agile practices such as sprint planning, daily stand-ups, sprint reviews, and retrospectives ensure better coordination among team members and faster issue resolution. The platform improves customer experience by offering 24/7 access, fast transaction processing, and real-time balance updates. It also benefits banks by automating record-keeping, reducing manual errors, and increasing operational efficiency. This project demonstrates how Agile methodology helps in developing high-quality, secure, and user-centric software solutions.

2. Introduction

2.1 Introduction

An Automate Banking Website enables customers to perform banking operations online. Agile methodology is used to ensure flexibility, security, and faster development.

2.2 Problem Identification

Traditional banking systems lack convenience, require physical visits, and suffer from slow manual processing.

2.3 Need of the Project

A modern platform is required for secure, instant, and 24/7 banking services.

2.4 Project Scheduling

Phase	Duration
Planning	2 Days
Development	8 Days
Testing	3 Days
Review	2 Days
Documentation	1 Day

2.5 Objectives

- Provide online fund management.
- Improve user security and experience.
- Use Agile methodology.
- Ensure secure transactions.
- Enable real-time balance checks.
- Enhance customer satisfaction.

3. Software Requirement Specification (SRS)

3.1 Purpose

To develop a secure and user-friendly Banking Website using Agile methodology.

3.2 Scope

Useful for banks, financial institutions, and account holders.

3.3 Hardware / Software Requirement

Hardware:

- 4GB RAM
- Intel i3 or above
- 500GB HDD

Software:

- Windows 10
- Web Technologies / Java
- MySQL
- VS Code / Eclipse

3.4 Tools

- HTML, CSS, JavaScript
- Java / JSP / Servlets
- MySQL
- Git
- Agile Tools (Jira / Trello)

3.5 Software Process Model

- **Agile Model**
- Sprint Planning
- Daily Stand-up
- Sprint Review
- Sprint Retrospective

4. System Design

4.1 Data Dictionary

Field	**Type**	**Description**
AccountNo	int	Unique Account Number
UserID	int	Unique User ID
CustomerName	String	Name of account holder
Balance	double	Current Account Balance
TransactionID	int	Unique Transaction ID
Type	String	Deposit/Withdraw/Transfer

4.2 ER Diagram

User → Account → Transaction

4.3 DFD

User → Banking Website → Database → Confirmation

4.4 Diagrams

- Use Case Diagram
- Activity Diagram
- Flowchart

5. Implementation

5.1 Program Code

Modules developed in sprints:

- User Registration & Login
- Account Dashboard
- Fund Transfer
- Deposit & Withdrawal
- Admin Panel

5.2 Output Screens

- Login Page
- Account Summary
- Transfer Funds Page
- Transaction Success Screen
- Transaction History

6. Testing

6.1 Test Data

Input	**Expected Output**
Invalid PIN	Login Failed / Access Denied
Empty Fields	Validation Error
Insufficient Funds	Transaction Failed Warning

6.2 Test Result

All features worked correctly after sprint-based testing.

7. User Manual

7.1 How to Use

1. Register/Login
2. View Account Balance
3. Select Transaction (Deposit/Withdraw)
4. Enter Amount and Confirm
5. View Transaction Receipt

7.2 Screen Layout

Web-based interface with easy navigation.

8. Project Applications & Limitations

8.1 Applications

- Online Banking
- Financial Management
- Fund Transfers
- Digital Payments

8.2 Limitations

- Internet dependency
- Limited offline support
- Server maintenance downtime

9. Conclusion & Future Enhancement

Conclusion:
The Automate Banking Website developed using Agile methodology provides a secure, flexible, and efficient solution for digital finance. Agile ensures faster delivery, better quality, and continuous improvement.
Future Enhancements:

- AI fraud detection
- Mobile app version
- Cloud hosting
- Chatbot support

10. Bibliography & References

- Agile Manifesto
- Scrum Guide
- Banking System Documentation
- GeeksForGeeks
- IEEE Journals